

Wave Equation

Let u is a dependent variable and x, y, z, t are independent variables then learn

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

One dimensional Wave equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

Two dimensional wave Equation

Three dimensional Wave equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

Here ' u ' is a solution of wave equation

Learn

Here we solve one dimension wave equation by method of separation of variables

Learn With using the conditions

(Conditions in two set)

① Boundary Conditions

(i) $u = 0$ when $x = 0$; $u(x, t) = u(0, t) = 0$

(ii) $u = 0$ when $x = l$
 $u(x, t) = u(l, t) = 0$

II. Initial Conditions

(i) $\frac{\partial u}{\partial t} = 0$ when $t = 0$
 $\left(\frac{\partial u}{\partial t} \right)_{t=0} = 0$

(ii) $u = f(x)$ when $t = 0$

VV Imp

Q. find the solution of one dimension wave equation

OR Solve

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

using method of separation of variables.

VV Imp

Q. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form

$$y = A \sin \frac{\pi x}{L}$$

from which it is released
at time $t=0$.

Show that the displacement
of any point at a distance
 x from one end at time
 t is given by

$$y(x, t) = A \sin \frac{\pi x}{L} \cos \frac{\pi c t}{L}$$

$$u(x, t) = u = y(x, t)$$

V.V.Imp Show how the wave

equation $c^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}$ can be

solved by the method of separation
of variables. If the initial

displacement and velocity of a string stretched between $x=0$ and $x=l$ are given by

$$y = f(x) \quad \text{and} \quad \frac{\partial y}{\partial t} = g(x),$$

determine the constants in the series solution

→ 12 questions in 15 min

Solution 4/1

i.e. solution of wave equation

Solution of one dimensional wave equation

By formula of one dimensional wave equation

Solution of One dimensional

Wave Equation

By formula of one dimensional wave equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} \quad \text{--- (1)}$$

By method of separation of variables

Let our solution is

$$u = XT \quad \text{--- (2)}$$

here ' x ' is the function of ' x ' only.

and ' T ' is the function of ' t ' only.

Since u is a solution, so

Satisfy equation (1) put the value

$$\frac{\partial^2}{\partial x^2} XT = \frac{1}{c^2} \frac{\partial^2 (XT)}{\partial t^2}$$

$$T \frac{\partial^2 X}{\partial x^2} = \frac{1}{c^2} X \frac{\partial^2 T}{\partial t^2}$$

$$T X'' = \frac{1}{c^2} X T''$$

Variable separable

Separate variable one side X

and other side T

$$T X'' = \frac{1}{c^2} X T''$$

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T}$$

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T}$$

Imp learn.
Value may be

+ve or -ve or zero

+p² or -p² or 0

We solve all three cases

Remember

Case I

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T} = +p^2 \text{ (let)}$$

$$\frac{X''}{X} = p^2$$

$$X'' = p^2 X$$

$$X'' - p^2 X = 0$$

Convert in 'D' form

$$\frac{1}{c^2} \frac{T'}{T} = p^2$$

$$T'' = c^2 p^2 T$$

$$T'' - c^2 p^2 T = 0$$

$$(D^2 - c^2 p^2) T = 0$$

$$(D^2 - p^2) X = 0$$

Solve by CF & P.I. method

Auxilliary equation

put $D=m$ equate to zero

$$m^2 - p^2 = 0$$

$$m^2 = p^2$$

$$m = +p, -p$$

$$C.f = C_1 e^{px} + C_2 e^{-px}$$

PT = 0
hence

$$X = C.f + P.I$$

$$X = C_1 e^{px} + C_2 e^{-px}$$

Solve by CF & P.I method

Auxiliary equation

put $D=m$ equate

to zero

$$m^2 - c^2 p^2 = 0$$

$$m^2 = c^2 p^2$$

$$m = +cp, -cp$$

$$C.f = C_3 e^{cpt} + C_4 e^{-cpt}$$

PT = 0
hence

$$T = C.f + P.I$$

$$T = C_3 e^{cpt} + C_4 e^{-cpt}$$

put value in (2)

In this case our wave equation solution is

$$u = X \cdot T$$

$$u = \left(c_1 e^{px} + c_2 e^{-px} \right) \cdot \left(c_3 e^{cpt} + c_4 e^{-cpt} \right) \quad \text{--- (3)}$$

Case II

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = -p \quad (\text{let})$$

$$\frac{X''}{X} = -p^2$$

$$X'' + p^2 X = 0$$

by Cof & P.I method

$$(D^2 + p^2) X = 0$$

put $D = m$

$$\frac{1}{c^2} \frac{T''}{T} = -p^2$$

$$T'' = -c^2 p^2 T$$

by Cof & P.I method

$$(D^2 + c^2 p^2) T = 0$$

$$A. \epsilon. \quad m^2 + p^2 = 0$$

$$m^2 = -p^2$$

$$m = \pm i p$$

$$m = \pm i p$$

$$C \cdot f = C_1 \cos px +$$

$$C_2 \sin px$$

$$P \cdot T = 0$$

$$X = C_1 \cos px +$$

$$C_2 \sin px$$

$$X = C_1 \cos px +$$

$$C_2 \sin px$$

$$A. \epsilon.$$

$$m^2 + c^2 p^2 = 0$$

$$m^2 = -c^2 p^2$$

$$m = \pm i c p$$

$$m = \pm i c p$$

$$C \cdot f = C_3 \cos cpt +$$

$$C_4 \sin cpt$$

$$P \cdot T = 0$$

$$T = C_3 \cos cpt +$$

$$C_4 \sin cpt$$

$$T = C_3 \cos cpt +$$

$$C_4 \sin cpt$$

$$u = X T$$

In this case our wave equation solution is

$$u = XT$$

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (4)}$$

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (4)}$$

Case III $\frac{X'}{X} = \frac{1}{c^2} \frac{T''}{T} = 0$ (let)

$$\frac{X''}{X} = 0$$

$$X'' = 0$$

$$D^2 X = 0$$

$$A.E. \quad m^2 = 0$$

$m = 0, 0$
repeat roots

$$\frac{1}{c^2} \frac{T''}{T} = 0$$

$$T'' = 0$$

$$D^2 T = 0$$

by CF and PI method

$$CF = C_1 + x C_2$$

$$PI = 0$$

$$X = C_1 + x C_2$$

Straight line

$$AE \quad m^2 = 0$$

$$m = 0, 0$$

$$CF = C_3 + t C_4$$

$$PI = 0$$

$$T = C_3 + t C_4$$

put values of X and T in

equation (2). In this case our

wave equation solution is

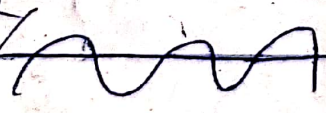
$$u = XT$$

$$u = (C_1 + x C_2) \cdot (C_3 + t C_4) \quad \text{--- (5)}$$

Of these three solutions we have to choose or select that solution which is consistent with the physical nature of the problem.

Since we are dealing with a problem on vibrations, so ' u ' must be a periodic function of x and t .

Therefore, the solution must involve trigonometric terms.

II case  sin
~~IV case~~ ~~exp~~ ~~I case~~ ~~exp.~~

Hence case II for $-p^2$ is only suitable solution of wave equation.

Ans

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (6)}$$

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (6)}$$

Ans

This is a solution of wave equation

Now in this solution we apply conditions for find the value of constants.

We apply two types of conditions

Boundary Conditions:

(i) $u = 0$ when $x = 0$

(ii) $u = 0$ when $x = l$
(string at length)

Initial Conditions

(iii) $\frac{\partial u}{\partial t} = 0$ when $t = 0$

(iv) $u = f(x)$ when $t = 0$

put condition in equation (6)

$$u=0 \text{ when } x=0$$

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 \cos cpt + C_4 \sin cpt)$$

$$0 = (C_1 \cdot 1 + C_2 \cdot 0) (C_3 \cos cpt + C_4 \sin cpt)$$

$$0 = C_1 \cdot (C_3 \cos cpt + C_4 \sin cpt)$$

not zero ~~if zero not possible~~
 ~~$\frac{\pi}{2}$~~

$$\boxed{C_1 = 0}$$

put condition (ii) in equation (6)
 $u=0$ when $x=l$

$$0 = (0 \cdot \cos pl + C_2 \sin pl) \cdot (C_3 \cos cpt + C_4 \sin cpt)$$

$$0 = \underline{C_2 \sin pl} \cdot (C_3 \cos cpt + C_4 \sin cpt)$$

not zero

$\frac{\pi}{2}$ zero

← not zero →

since $C_1=0$
if $C_2=0$ then
 $\frac{\pi}{2}$ $x=0$ zero

so $\sin pl = 0$

$$\sin pl = \sin n\pi$$

$pl = \frac{n\pi}{l}$

$l \rightarrow$ length of string

put these values in eqⁿ (6)
we get wave equation solution
in another form

$$u = \left(0 + \sum_2 \frac{\sin \frac{n\pi x}{l}}{l} \right) \cdot \left(\sum_3 \frac{\cos \frac{cn\pi t}{l}}{l} + \sum_4 \frac{\sin \frac{cn\pi t}{l}}{l} \right)$$

$$u = \left(\sum_2 \sum_3 \frac{\cos \frac{n\pi ct}{l}}{l} + \sum_2 \sum_4 \frac{\sin \frac{n\pi ct}{l}}{l} \right) \sin \frac{n\pi x}{l}$$

let replace constant $\sum_2 \sum_3$ by

a_n or more general as a_n

and $C_2 C_4$ by b_n or more
general b_n

i.e. let $C_2 C_3 = a_n$ and $C_2 C_4 = b_n$

we get

$$u = \left(a_n \frac{\cos n\pi ct}{l} + b_n \frac{\sin n\pi ct}{l} \right) \frac{\sin n\pi x}{l}$$

This is also wave eqⁿ solution 7

put initial condition (iii) in eqn 7

$$\frac{\partial u}{\partial t} = 0 \quad \text{when } t = 0$$

$$\frac{\partial u}{\partial t} = \left(a_n \left(\frac{-\sin n\pi ct}{l} \right) \frac{n\pi c}{l} + b_n \frac{\cos n\pi ct}{l} \cdot \frac{n\pi c}{l} \right) \frac{\sin n\pi x}{l}$$

$$0 = \left(0 + b_n \times 1 \cdot \frac{n\pi c}{l} \right) \frac{\sin n\pi x}{l}$$

$$0 = b_n \frac{n\pi c}{l} \sin \frac{n\pi x}{l}$$

$$0 = b_n \frac{\sin n\pi x}{l}$$

$$b_n = 0$$

put this value in equation 7

$$u = a_n \frac{\cos n\pi ct}{l} \frac{\sin n\pi x}{l} + 0 \cdot \frac{\sin n\pi ct}{l}$$

$$u = a_n \frac{\cos n\pi ct}{l} \cdot \frac{\sin n\pi x}{l}$$

Learn as Result

This is final solution of
One dimension wave equation

Also the most general solution of wave equation in one dimension is

$$u = \sum_{n=1}^{\infty} a_n \cos \frac{n\pi ct}{L} \cdot \sin \frac{n\pi x}{L}$$

learn as Result — (9)

We use direct in many questions

Put initial condition (iv) in equation

(9) $u = f(x)$ when $t=0$

$$f(x) = \sum_{n=1}^{\infty} a_n \cos 0 \cdot \sin \frac{n\pi x}{L}$$

$$f(x) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi x}{L} \quad \text{--- (10)}$$

This equation (10) represents Fourier series for $f(x)$

where the value of a_n
given by fourier series

formula

$$a_n = \frac{2}{L} \int_0^L \frac{\sin n\pi x}{L} f(x) dx$$

(11)

and $b_n = 0$ for all n — (12)

* Equation (8) and (9) are
final solution of one
Dimension Wave Equation.

OR Question

Do all as previous

Solution of wave equation i.e.
eqn (8)

$$u = a_n \cos \frac{n\pi ct}{L} \sin \frac{n\pi x}{L}$$

put $n=1$

$$u = a_1 \cos \frac{\pi ct}{L} \sin \frac{\pi x}{L}$$

Replace constant a_1 by A

and write $u = u(x, t) = y(x, t)$

We get

$$y(x, t) = A \cos \frac{\pi ct}{L} \sin \frac{\pi x}{L}$$

Ans

OR Question All are same as
but initial condition is changed in
this question (iii) $\frac{\partial y}{\partial t} = g(x)$ when $t=0$

(iv) $u = f(x)$ when $t=0$

by equation (7)

$$u = \left(a_n \cos \frac{n\pi ct}{L} + b_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}$$

put condition (iii)

$$\frac{\partial u}{\partial t} = \left[a_n \left(-\sin \frac{n\pi ct}{L} \right) \frac{n\pi c}{L} + \right.$$

$$\left. b_n \cos \frac{n\pi ct}{L} \cdot \frac{n\pi c}{L} \right] \sin \frac{n\pi x}{L}$$

put $t=0$

$$g(x) = \left[0 + b_n \frac{n\pi c}{L} \right] \sin \frac{n\pi x}{L}$$

$$g(x) = b_n \frac{n\pi c}{L} \sin \frac{n\pi x}{L}$$

$$g(x) = \frac{n\pi c}{L} b_n \sin \frac{n\pi x}{L}$$

This is Fourier expansion of $g(x)$

and value of constant b_n is

$$b_n = \frac{L}{n\pi c} \int_0^L g(x) \sin \frac{n\pi x}{L}$$

$$b_n = \frac{2}{n\pi c} \int_0^L g(x) \sin \frac{n\pi x}{L} \quad \text{***}$$

put condition (iv) in (i) equation

$$f(x) = \left(a_n \cdot 0 + b_n \sin \frac{n\pi x}{L} \right)$$

$$f(x) = \sum a_n \sin \frac{n\pi x}{L}$$

This is fourier expansion of $f(x)$ and value of constant a_n is

$$a_n = \frac{2}{L} \int_0^L \sin \frac{n\pi x}{L} f(x) dx$$

Hence equation (*) is solution of wave equation and the value of constant is given by equation ** and ***

Hence